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FET and MOSFET

GATE ECE Analog Circuits · Previous Year Questions with Answer Key

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Marks 2	19 questions
Marks 5	3 questions
Marks 8	1 question
Answer key	at the end

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Marks 1 10 questions

Q1. GATE ECE 2025

Which of the following statements is/are TRUE with respect to ideal MOSFET-based DCcoupled single-stage amplifiers having finite load resistors?

Open solution

(A) The common-gate amplifier has an infinite input resistance

(B) The common-source amplifier has an infinite input resistance

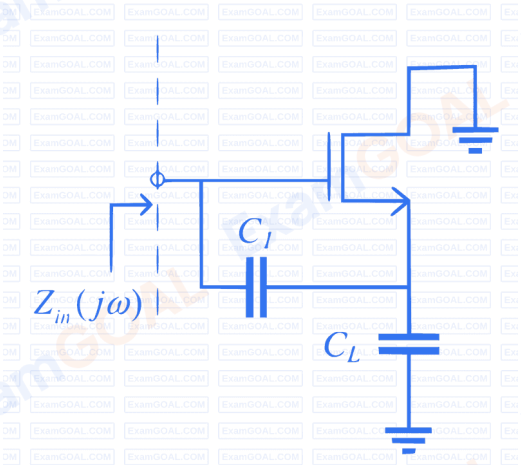
(C) The input and output voltages of the common-source amplifier are in phase

(D) The input and output voltages of the common-drain amplifier are in phase

Q2. GATE ECE 2024

In the circuit below, assume that the long channel NMOS transistor is biased in saturation. The small signal trans-conductance of the transistor is g_m. Neglect body effect, channel length modulation, and intrinsic device capacitances. The small signal input impedance $Z_{in}(j\omega)$ is _____

Open solution



(A) $\frac{-g_m}{C_1C_L\omega^2} + \frac{1}{j\omega C_1} + \frac{1}{j\omega C_L}$

(B) $\frac{g_m}{C_1C_L\omega^2} + \frac{1}{j\omega C_1} + \frac{1}{j\omega C_L}$

(C) $\frac{1}{j\omega C_1} + \frac{1}{j\omega C_L}$

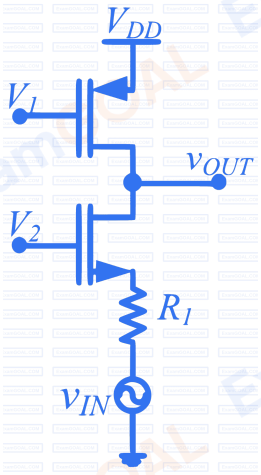
(D) $\frac{-g_m}{C_1C_L\omega^2} + \frac{1}{j\omega C_1 + j\omega C_L}$

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Q3. GATE ECE 2023

Open solution

In the circuit shown below, V_1 and V_2 are bias voltages. Based on input and output impedances, the circuit behaves as a

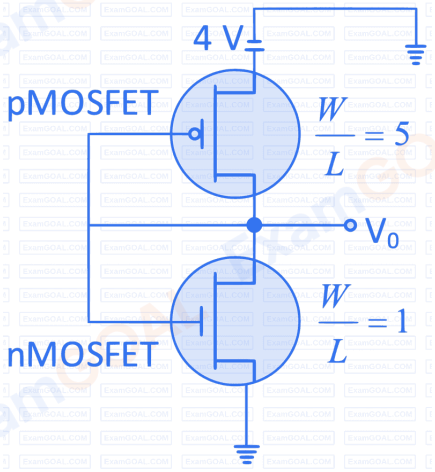


(A) voltage controlled voltage source.
(B) voltage controlled current source.
(C) current controlled voltage source.
(D) current controlled current source.

Q4. GATE ECE 2022

Open solution

Consider the CMOS circuit shown in the figure (substrates are connected to their respective sources). The gate width (W) to gate length (L) ratios $\left(\frac{W}{L}\right)$ of the transistors are as shown. Both the transistors have the same gate oxide capacitance per unit area. For the pMOSFET, the threshold voltage is -1 V and the mobility of holes is $40\frac{cm^2}{V.s}$. For the nMOSFET, the threshold voltage is 1 V and the mobility of electrons is $300\frac{cm^2}{V.s}$. The steady state output voltage V_0 is _____.



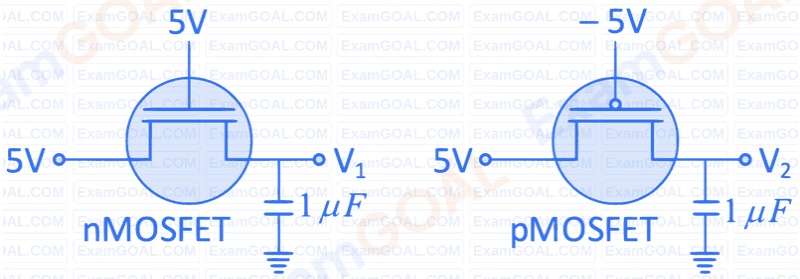
(A) equal to 0 V
(B) more than 2 V
(C) less than 2 V
(D) equal to 2 V

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Q5. GATE ECE 2022

Open solution

The ideal long channel nMOSFET and pMOSFET devices shown in the circuits have threshold voltages of 1 V and -1 V, respectively. The MOSFET substrates are connected to their respectively sources. Ignore leakage currents and assume that the capacitors are initially discharged. For the applied voltages as shown, the steady state voltages are _____.



(A) V_1 = 5 V, V_2 = 5 V

(B) V_1 = 5 V, V_2 = 4 V

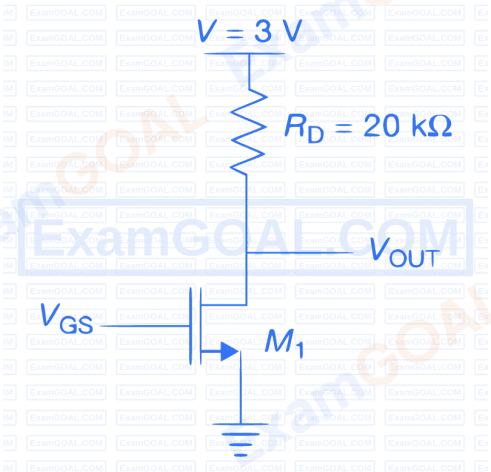
(C) V_1 = 4 V, V_2 = 5 V

(D) V_1 = 4 V, V_2 = -5 V

Q6. GATE ECE 2021

Open solution

For the transistor M_1 in the circuit shown in the figure, $\mu_n C_{ox} = 100\mu\ \mathrm{A}/V^2$ and $\frac{W}{L} = 10$, where μ_n is the mobility of electron, C_{ox} is the oxide capacitance per unit area. W is the width and L is the length.



The channel length modulation coefficient is ignored. If the gate-to-source voltage V_{GS} is 1 V to keep the transistor at the edge of saturation. Then the threshold voltage of the transistor (rounded off to one decimal place) is ____ V.

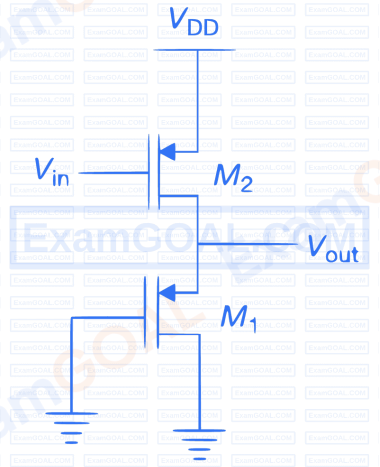
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Q7. GATE ECE 2021

Open solution

In the circuit shown in the figure, the transistors M_1 and M_2 are operating in saturation. The channel length

modulation coefficients of both the transistors are non-zero. The transconductance of the MOSFETs M_1 and M_2 are g_{m1} and g_{m2}, respectively, and the internal resistance of the MOSFETs M_1 and M_2 are r_{01} and r_{02} respectively.



Ignoring the body effect, the ac small signal voltage gain (dV_{out}/dV_{in}) of the circuit is

(A) $-g_{m2}\,(r_{01} \amalg r_{02})$

(B) $-g_{m2}\left(\frac{1}{g_{m1}} \| r_{02}\right)$

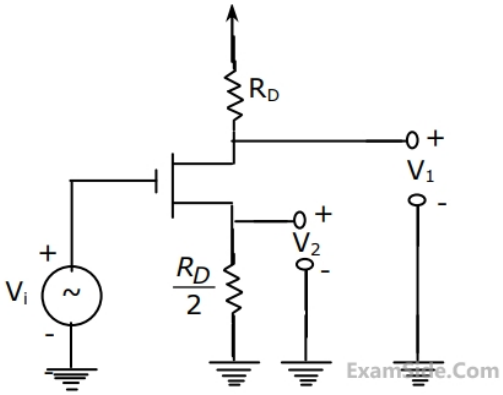
(C) $-g_{m1}\left(\frac{1}{g_{m2}} \| r_{01} \| r_{02}\right)$

(D) $-g_{m2}\left(\frac{1}{g_{m1}} \| r_{01} \| r_{02}\right)$

Q8. GATE ECE 1998

Open solution

In the MOSFET amplifier of the figure the signal output V_1 and V_2 obey the relationship



(A) $V_1 = \frac{V_2}{2}$

(B) $V_1 = -\frac{V_2}{2}$

(C) $V_1 = 2V_2$

(D) $V_1 = -2V_2$

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Q9. GATE ECE 1995

An n-channel JFET has I_{DSS} = 1 mA and V_p = -5 V. Its maximum transconductance is _____

Open solution

Q10. GATE ECE 1994

The transit time of a current carriers through the channel of an FET decides its _____characteristics.

Open solution

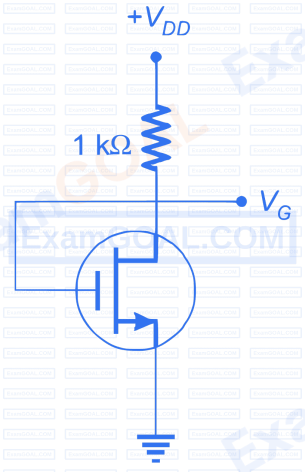
Marks 2 19 questions

Q11. GATE ECE 2026

An n-channel MOSFET is connected as shown in the Figure.

Assume $\mathrm{V_{TH}} = 1\ \mathrm{V}, V_{DD} = 5\ \mathrm{V}$, and $\mu C_{Ox}\left(\frac{W}{L}\right) = 2\mathrm{mAV}^{-2}$ and neglect channel length modulation effects.

The gate voltage (V_G) of the n-channel MOSFET (in Volt) is ____ . (rounded off to two decimal places)



Open solution

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Q12. GATE ECE 2025

Open solution

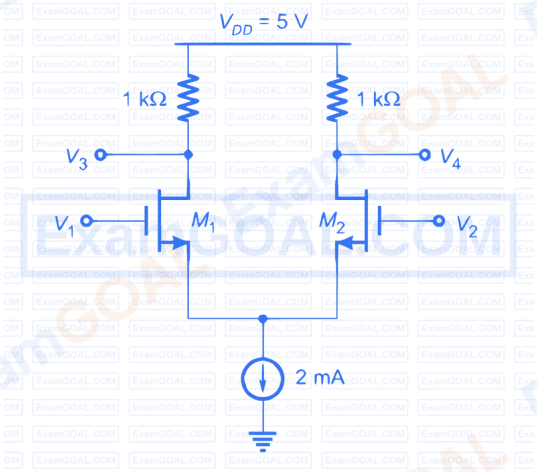
The identical MOSFETs M_1 and M_2 in the circuit given below are ideal and biased in the saturation region. M_1 and M_2 have a transconductance g_m of 5 mS .

The input signals (in Volts) are:

$V_1 = 2.5 + 0.01 \sin \omega t$

$V_2 = 2.5 - 0.01 \sin \omega t$

The output signal V_3 (in Volts) is _ .



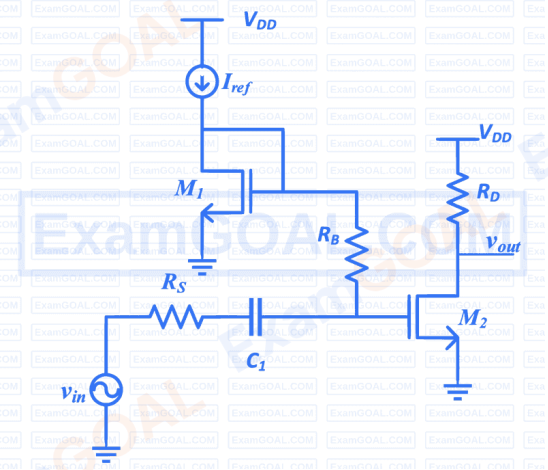
(A) $3 + 0.05 \sin \omega t$ (B) $3 - 0.1 \sin \omega t$ (C) $4 + 0.1 \sin \omega t$ (D) $4 - 0.05 \sin \omega t$

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Q13. GATE ECE 2024

Open solution

In the circuit shown below, the transistors M_1 and M_2 are biased in saturation. Their small signal transconductances are g_{m1} and g_{m2} respectively. Neglect body effect, channel length modulation and intrinsic device capacitances.



Assuming that capacitor C_i is a short circuit for AC analysis, the exact magnitude of small signal voltage gain $\left|\frac{v_{out}}{v_{in}}\right|$ is _____.

(A) $g_{m2}R_D$

(B) $\frac{g_{m2}R_D\left(R_B+\frac{1}{g_{m1}}\right)}{R_B+\frac{1}{g_{m1}}+R_S}$

(C) $\frac{g_{m2}R_D\left(R_B+\frac{1}{g_{m1}}+R_S\right)}{R_B+\frac{1}{g_{m1}}}$

(D) $\frac{g_{m2}R_D\left(\frac{1}{g_{m1}}\right)}{\frac{1}{g_{m1}}+R_S}$

Q14. GATE ECE 2024

Open solution

An NMOS transistor operating in the linear region has I_D of 5 μA at V_{DS} of 0.1 V. Keeping V_{GS} constant, the V_{DS} is increased to 1.5 V.

Given that $\mu_n C_{ox}\frac{W}{L}$ = 50 μA/V^2, the transconductance at the new operating point (in μA/V, *rounded off to two decimal places*) is _____.

Q15. GATE ECE 2022

Open solution

Consider an ideal long channel nMOSFET (enhancement-mode) with gate length 10 μm and width 100 μm. The product of electron mobility (μ_n) and oxide capacitance per unit area (C_{ox}) is $\mu_n C_{ox}$ = 1 mA/V^2. The threshold voltage of the transistor is 1 V. For a gate-to-source voltage V_{GS} = [2 − sin(2t)] V and drain-to source voltage V_{DS} = 1 V (substrate connected to the source), the maximum value of the drain-to-source current is _____.

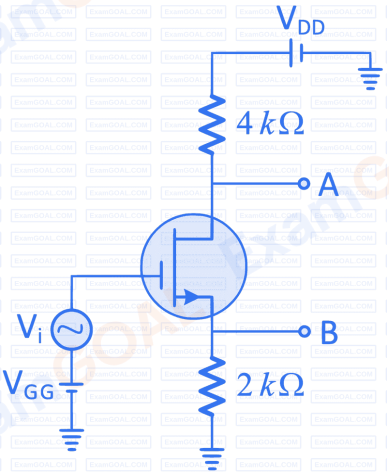
(A) 40 mA (B) 20 mA (C) 15 mA (D) 5 mA

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Q16. GATE ECE 2022

Open solution

Consider the circuit shown with an ideal long channel nMOSFET (enhancement mode, substrate is connected to the source). The transistor is appropriately biased in the saturation region with V_{GG} and V_{DD} such that it acts as a linear amplifier. v_i is the small-signal ac input voltage. v_A and v_B represent the small-signal voltages at the nodes A and B, respectively. The value of $\frac{v_A}{v_B}$ is _____ (rounded off to one decimal place).



Q17. GATE ECE 2021

Open solution

For an n-channel silicon MOSFET with 10 nm gate oxide thickness, the substrate sensitivity ($\partial V_T / \partial |V_{BS}|$) is found to be 50mV/V at a substrate voltage $|V_{BS}| = 2$ V, where V_T is the threshold voltage of the MOSFET. Assume that, $|V_{BS}| \gg 2\phi_B$, where $q\phi_B$ is the separation between the Femi energy level E_F and the intrinsic level E_i in the bulk. Parameters given are

Electron charge $(q) = 1.6 \times 10^{-9}\text{C}$

Vacuum permittivity $(\varepsilon_o) = 8.85 \times 10^{-12}\ \text{F/m}$

Relative permittivity of silicon $(\varepsilon_{Si}) = 12$

Relative permittivity of oxide $(\varepsilon_{ox}) = 4$

The doping concentration of the substrate is

(A) $2.37 \times 10^{15}\ \text{cm}^{-3}$

(B) $4.37 \times 10^{15}\ \text{cm}^{-3}$

(C) $9.37 \times 10^{15}\ \text{cm}^{-3}$

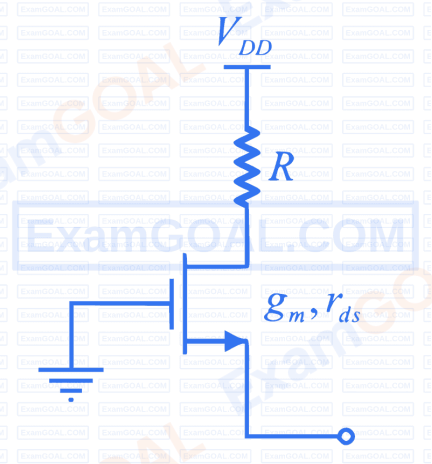
(D) $7.37 \times 10^{15}\ \text{cm}^{-3}$

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Q18. GATE ECE 2020

Open solution

Using the incremental low frequency small - signal model of the MOS device, the Norton equivalent resistance of the following circuit is

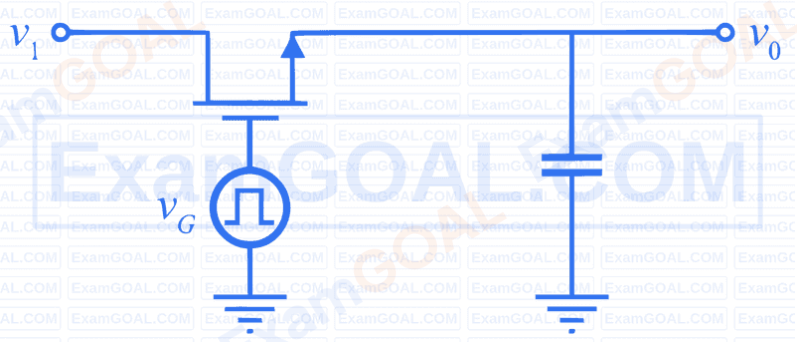


(A) $r_{ds} + R$ (B) $\frac{r_{ds}+R}{1+g_m r_{ds}}$ (C) $r_{ds} + R + g_m r_{ds} R$ (D) $r_{ds} + \frac{1}{g_m} + R$

Q19. GATE ECE 2020

Open solution

An enhancement MOSFET of threshold voltage 3 V is being used in the sample and hold circuit given below. Assume that the substrate of the MOS device is connected to -10 V . If the input voltage v_1 liesbetween ± 10 V, the minimum and the maximum value of v_G required for proper sampling and holding respectively, are



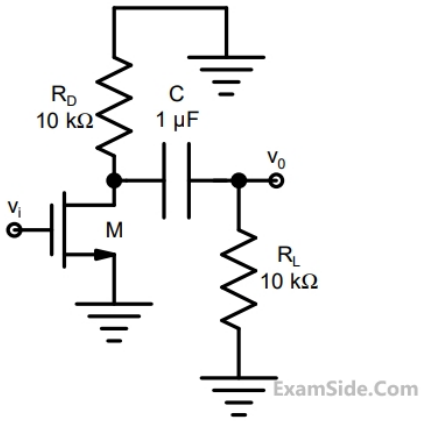
(A) 10 V and -10 V (B) 13 V and -7 V (C) 10 V and -13 V (D) 3 V and -3 V

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Q20. GATE ECE 2013

Open solution

The ac schematic of an NMOS common-source stage is shown in the figure below, where part of the biasing circuits has been omitted for simplicity. For the n -channel MOSFET M, the transconductance g_m = 1 mA/V, and body effect and channel length modulation effect are to be neglected. The lower cutoff frequency in Hz of the circuit is approximately at



(A) 8 (B) 32 (C) 50 (D) 200

Q21. GATE ECE 2006

Open solution

An n-channel depletion MOSFET has following two points on its I_D - V_{GS} curve:

(i)V_{GS} = 0 at I_d = 12 mA and

(ii)V_{GS} = -6 Volts at Z_o =∞

Which of the following Q-points will give the highest transconductance gain for small signals?

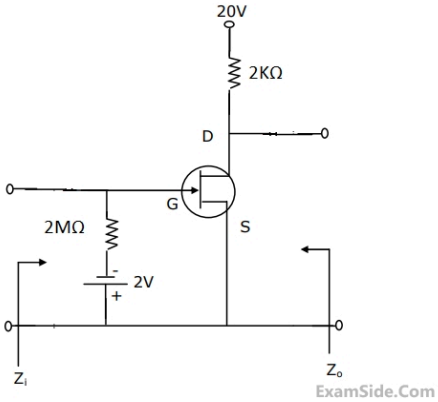
(A) $V_{GS} = -6\,Vots$ (B) $V_{GS} = -3\,Vots$ (C) $V_{GS} = 0\,Vots$ (D) $V_{GS} = 3\,Vots$

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Q22. GATE ECE 2005

Open solution

Given
$r_d = 20K\Omega,\ I_{DSS} = 10mA,\ V_P = -8V$



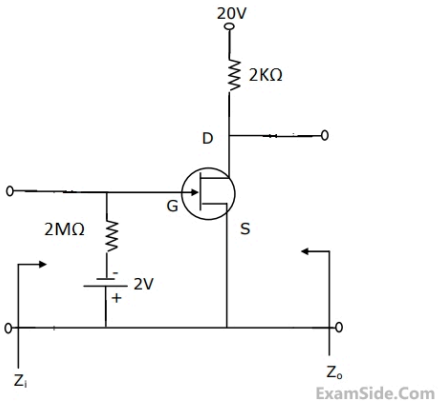
Z_i and Z_o of the circuit are respectively

(A) 2 $M\Omega$ and 2 $K\Omega$
(B) 2 $M\Omega$ and 20/11 $K\Omega$
(C) infinity and 2 $M\Omega$
(D) infinity and 20/11$M\Omega$

Q23. GATE ECE 2005

Open solution

Given
$r_d = 20K\Omega,\ I_{DSS} = 10mA,\ V_P = -8V$



I_D and V_{DS} under DC conditions are respectively

(A) 5.625 mA and 8.75 V
(B) 7.500 mA and 5.00V
(C) 4.500 mA and 11.00V
(D) 6.250 mA and 7.50V

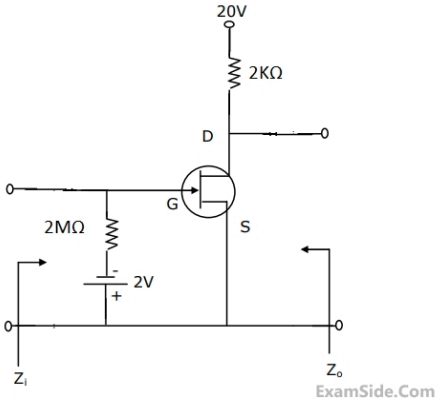
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Q24. GATE ECE 2005

Open solution

Given

$r_d = 20K\Omega,\ I_{DSS} = 10mA,\ V_P = -8V$



Transconductance in milli-Siemens (mS) and voltage gain of the amplifier are respectively.

(A) 1.875 mS and 3.41

(B) 1.875 mS and -3.41

(C) 3.3 mS and -6

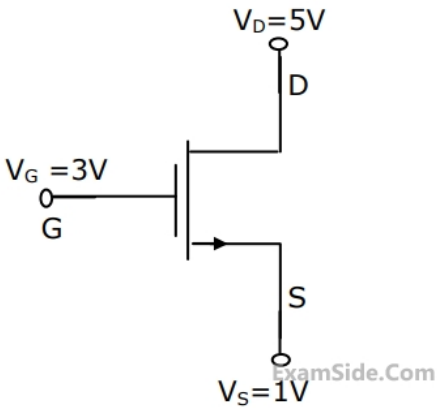
(D) 3.3 mS and 6

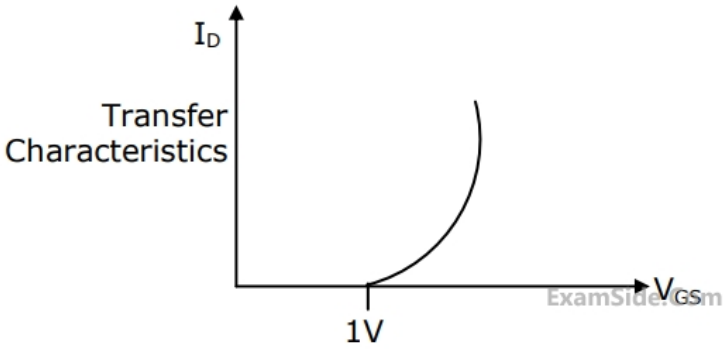
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Q25. GATE ECE 2005

Open solution

For an n-channel MOSFET and its transfer curve shown in the figure, the threshold voltage is





(A) 1 V and the device is in active region

(B) -1 V and the device is in saturation region.

(C) 1 V and the device is in saturation region.

(D) -1 V and the device is in active region.

Q26. GATE ECE 2003

Open solution

The action of a JFET in its equivalent circuit can be best represented as a

(A) current controlled current source

(B) current controlled voltage source

(C) voltage controlled voltage source

(D) voltage controlled current source

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Q27. GATE ECE 2002

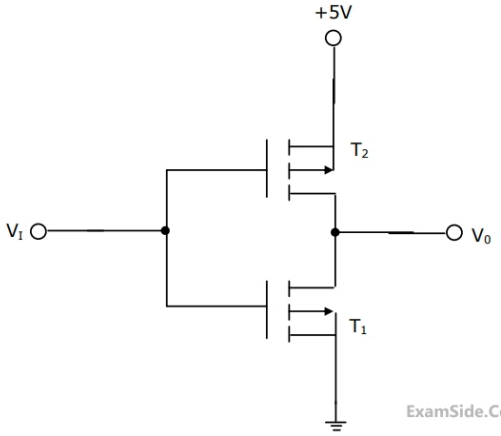
Open solution

Consider the following statements in connection with the CMOS inverter in the Figure. Where both the MOSFETS are of enhancement type and both have a threshold voltage of 2V.

Statement 1: T_1 conducts when $V_i \geq 2\,V$.

Statement 2: T_1 is always in saturation when $V_0 = 0\,V$.

Which of the following is correct?



(A) only Statement 1 is true

(B) Only Statement 2 is true

(C) Both the statements are true

(D) Both the statements are FALSE

Q28. GATE ECE 1992

Open solution

An n-channel JFET has a pinch-off voltage of V_p = -5V. V_{DS}(max) = 20V and g_m = 2mA/V. The minimum 'ON' resistance is achieved in the JFET for

(A) $V_{GS} = -7V$ and $V_{DS} = 0V$

(B) $V_{GS} = 7V$ and $V_{DS} = 0V$

(C) $V_{GS} = 0V$ and $V_{DS} = 20V$

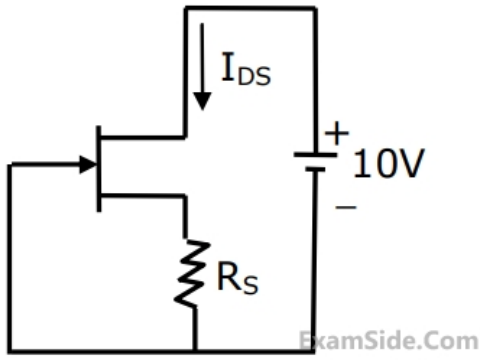
(D) $V_{GS} = -7V$ and $V_{DS} = 20V$

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Q29. GATE ECE 1992

The JFET in the circuit shown in fig. has an I_{DSS} = 10mA and V_p = -5V. The value of the resistance R_s for a drain current I_{DS} = 6.4mA is (select the Nearest value).

Open solution



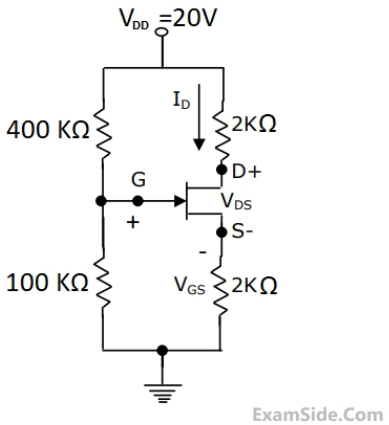
(A) 150 ohms (B) 470 ohms (C) 560 ohms (D) 1 Kilo ohm

Marks 5 3 questions

Q30. GATE ECE 1996

A JEFT with V_P = -4V and I_{Dss} = 12mA is used in the circuit shown in Fig. Assuming the device to be operating in saturation, determine I_D, V_{DS} and V_{GS}

Open solution

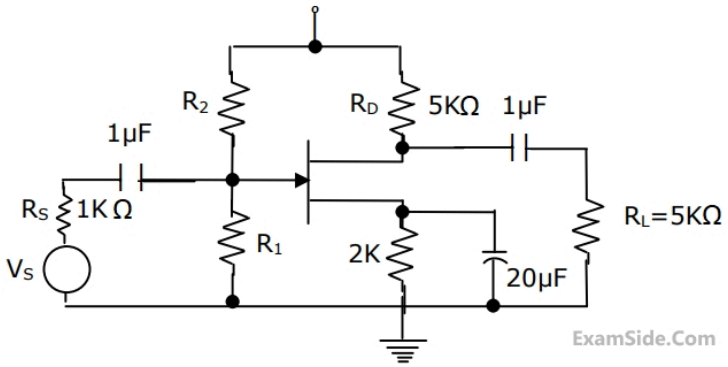


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Q31. GATE ECE 1995

In the JFET circuit shown in Fig assume that $R_1||R_2$ = 1 MΩ and the total stray capacitance at the output to be 20 pF. The JFET used has g_m = 2mA/V, C_{gs} = 20 pF and C_{gd} = 2 pF. Determine the upper cut-off frequency of the amplifier.

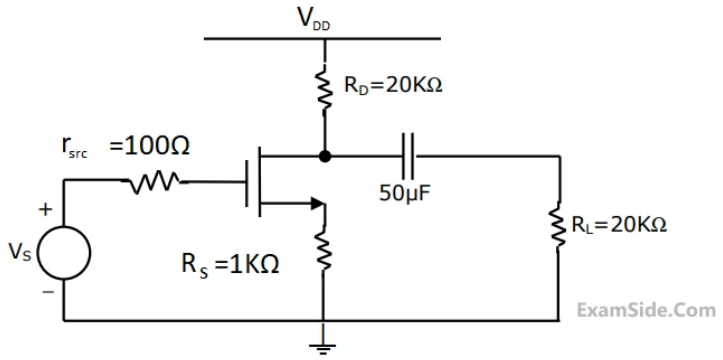
Open solution



Q32. GATE ECE 1994

In the MOSFET amplifier shown in the fig. below, the transistor has $\mu = 50,\ r_d = 10\,K\Omega,\ C_{gs} = 5\ pF, C_{gd} = 1pF$ and $C_{ds} = 2\,pF$. Draw a small signal equivalent circuit for the amplifier for midband frequencies and calculate its midband voltage gain.

Open solution



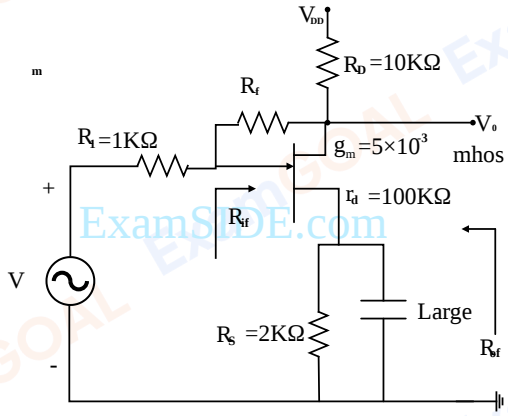
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Marks 8 1 questions

Q33. GATE ECE 1987

Open solution

(a) Specify the type of the negative feedback in the JFET amplifier shown in Fig.
(b)Calculate the voltage gain (V_0/V_i), of the amplifier with no feedback ($R_f=\infty$).
(c)Determine the value of the resistance R_f, required to give a voltage gain (V_0/V_i), or - 10.



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Answer Key

Marks 1

1. **B, D**	2. **A**	3. **D**	4. **C**	5. **C**
6. **0.5**	7. **D**	8. **D**	9. **0.4 ms**	10. **Switching**

Marks 2

11. **(2.56) (2.50 to 2.60)**	12. **D**	13. **B**	14. **52.40 to 52.60**	15. **C**
16. **-2**	17. **D**	18. **B**	19. **B**	20. **A**
21. **D**	22. **B**	23. **A**	24. **B**	25. **C**
26. **D**	27. **A**	28. **C**	29. **A**	

Marks 5

30. **I_D = 3 mA V_{DS} = 4 V V_{GS} = 2 V**	31. **upper cut-off frequency = 2.89 MHz**	32. **Ω Midband voltage gain = -8.3**

Marks 8

33. **(a) This is a voltage shunt feedback Amplifier. (b)voltage gain = -454 (c) R_f = 10 KΩ**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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